

Advanced Algebra & Systems



Systems of three linear equations

- 1 Solve the system: $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 5$.
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Solving polynomial equations of higher degree

- 2 Solve $x^3 - 6x^2 + 11x - 6 = 0$, given that $x = 1$ is a root.
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Rational root theorem

- 3 List the possible rational roots of $2x^3 - 3x^2 - 8x + 12 = 0$, using the rational root theorem, then find one actual root by testing.
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Partial fractions

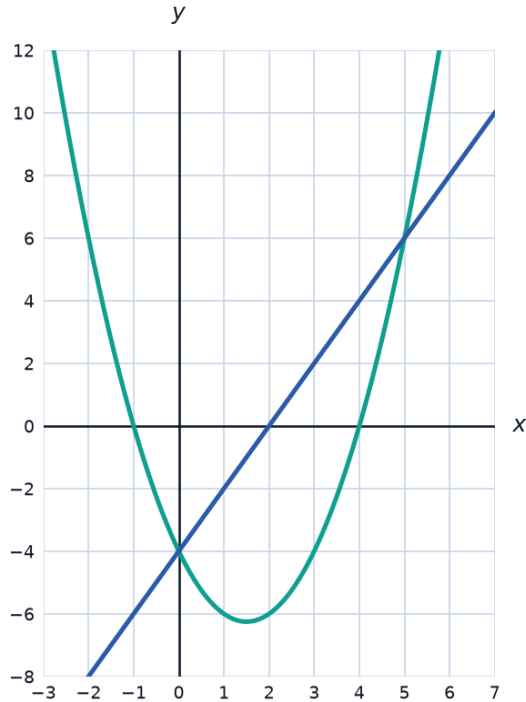
- 4 Express $\frac{5x - 1}{(x - 1)(x + 2)}$ in partial fractions.
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Radical equations with extraneous solutions

- 5 Solve $\sqrt{2x + 7} = x + 2$, and check for extraneous solutions.
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Systems with a quadratic and a linear equation

- 6 Solve the system: $y = x^2 - 3x - 4$ and $y = 2x - 4$.



Logarithmic and exponential equations

- 7 Solve $\log_2(x + 3) = 5$.
- 8 Solve $3^{2x-1} = 27$.

Inequalities involving polynomials

- 9 Solve $x^2 - x - 6 > 0$, giving your answer using interval notation reasoning.

A multi-part system and polynomial combination problem

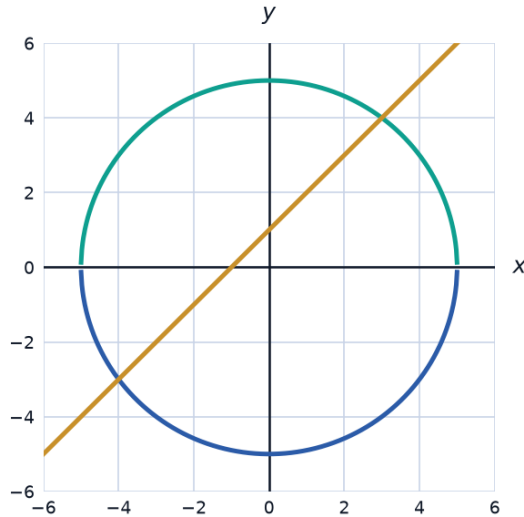
- 10 This question has two parts. A cubic polynomial $f(x) = x^3 + ax^2 + bx - 6$ has $x = 1$ and $x = 2$ as roots. (a) Use the factor theorem to set up and solve a system for a and b . (b) Find the third root of the polynomial.

Synthetic division

- 11 Use synthetic division to divide $2x^3 - 5x^2 + 3x - 1$ by $(x - 2)$, and state the quotient and remainder.

Nonlinear systems with three variables

- 12 Solve the system: $x^2 + y^2 = 25$ and $y = x + 1$.



Simplifying complex rational expressions

- 13 Simplify $\frac{\frac{1}{x} + \frac{1}{y}}{\frac{1}{x} - \frac{1}{y}}$.
- 14 Simplify $\frac{x^2 - 4}{x^2 - 5x + 6}$.

Solving equations with rational exponents

- 15 Solve $x^{2/3} = 9$ for x .

A multi-part rational function and system problem

- 16 This question has three parts. A boat's fuel efficiency is modeled by $E(v) = \frac{600}{v} + 0.02v^2$, where v is speed in mph and E is fuel used (gallons) for a fixed trip. (a) Find $E(20)$. (b) Find $E(30)$. (c) Based on parts (a) and (b), determine whether increasing speed from 20 to 30 mph increases or decreases fuel usage for this trip, and explain the two competing effects in the formula that cause this.

Solving a system involving a rational equation

- 17 Solve the system: $\frac{1}{x} + \frac{1}{y} = \frac{5}{6}$ and $x + y = 5$.
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Multiple choice — easy

- 18 Solve $\log_3(x + 4) = 4$.

a. 76
b. 78
c. 77
d. 81

- 19 Solve $\log_2(x + 18) = 4$.

a. 16
b. -3
c. -1
d. -2

- 20 Solve $\log_4(x - 10) = 4$.

a. 265
b. 266
c. 256
d. 267

- 21 Solve $\log_2(x + 15) = 2$.

a. 4
b. -12
c. -11
d. -10

22 Solve $\log_2(x + 14) = 5$.

a. 19

b. 32

c. 17

d. 18

23 Solve $\log_2(x + 8) = 5$.

a. 24

b. 25

c. 32

d. 23

24 Solve $\log_2(x) = 1$.

a. 3

b. 2

c. 1

d. 0

25 Solve $\log_3(x + 12) = 2$.

a. 9

b. -2

c. -3

d. -4

26 Solve $\log_2(x - 10) = 2$.

a. 14

b. 4

c. 13

d. 15

27 Solve $\log_2(x - 4) = 2$.

a. 8

b. 9

c. 4

d. 7

28 Solve $\log_4(x + 15) = 2$.

a. 2

b. 1

c. 16

d. 0

29 Solve $\log_3(x + 17) = 5$.

a. 226

b. 243

c. 225

d. 227

30 Solve $\log_3(x + 9) = 5$.

- a. 233
- b. 243
- c. 234
- d. 235

31 Solve $\log_4(x - 9) = 3$.

- a. 72
- b. 74
- c. 73
- d. 64

32 Solve $\log_3(x + 20) = 2$.

- a. -12
- b. 9
- c. -10
- d. -11

Multiple choice — medium

33 Solve $x^{2/3} = 484$ for x .

- a. 10648
- b. 10670
- c. 10626
- d. 484

34 Solve $x^{2/3} = 361$ for x .

- a. 361
- b. 6859
- c. 6840
- d. 6878

35 Solve $x^{2/3} = 9$ for x .

- a. 24
- b. 30
- c. 27
- d. 9

36 Solve $x^{2/3} = 49$ for x .

- a. 343
- b. 49
- c. 336
- d. 350

37 Solve $x^{2/3} = 256$ for x .

- a. 4112
- b. 256
- c. 4080
- d. 4096

38 Solve $x^{2/3} = 36$ for x .

a. 222

b. 36

c. 210

d. 216

39 Solve $x^{2/3} = 529$ for x .

a. 12167

b. 12190

c. 529

d. 12144

40 Solve $x^{2/3} = 625$ for x .

a. 625

b. 15625

c. 15600

d. 15650

41 Solve $x^{2/3} = 289$ for x .

a. 4913

b. 4896

c. 289

d. 4930

42 Solve $x^{2/3} = 100$ for x .

a. 1000

b. 1010

c. 100

d. 990

43 Solve $x^{2/3} = 64$ for x .

a. 512

b. 520

c. 504

d. 64

44 Solve $x^{2/3} = 4$ for x .

a. 8

b. 4

c. 10

d. 6

45 Solve $x^{2/3} = 169$ for x .

a. 2184

b. 169

c. 2210

d. 2197

46 Solve $x^{2/3} = 25$ for x .

- a. 25
- b. 125
- c. 130
- d. 120

47 Solve $x^{2/3} = 576$ for x .

- a. 13824
- b. 576
- c. 13800
- d. 13848

Multiple choice — hard

48 Solve $3^{3x+2} = 243$ for x .

- a. 3
- b. 0
- c. 2
- d. 1

49 Solve $5^{4x} = 625$ for x .

- a. 1
- b. 2
- c. 0
- d. 3

50 Solve $5^{2x+1} = 78125$ for x .

a. 4

b. 5

c. 3

d. 2

51 Solve $2^{3x-5} = 128$ for x .

a. 4

b. 5

c. 6

d. 3

52 Solve $5^{x+8} = 15625$ for x .

a. -3

b. -2

c. -1

d. 0

53 Solve $x^3 + x^2 - 50x + 48 = 0$, given that $x = 1$ is a root. Find the other two roots as an ordered pair (smaller first).

a. $(8, -6)$

b. $(-8, 6)$

c. $(-8, 5)$

d. $(-7, 6)$

- 54 Solve $x^3 - 36x = 0$, given that $x = 6$ is a root. Find the other two roots as an ordered pair (smaller first).
- a. $(6, 0)$
 - b. $(-6, -1)$
 - c. $(-5, 0)$
 - d. $(-6, 0)$
- 55 Solve $x^3 - 3x^2 - 18x = 0$, given that $x = 6$ is a root. Find the other two roots as an ordered pair (smaller first).
- a. $(-2, 0)$
 - b. $(-3, 0)$
 - c. $(-3, -1)$
 - d. $(3, 0)$
- 56 Solve $x^3 - 6x^2 - 7x + 60 = 0$, given that $x = 4$ is a root. Find the other two roots as an ordered pair (smaller first).
- a. $(-2, 5)$
 - b. $(-3, 5)$
 - c. $(3, -5)$
 - d. $(-3, 4)$
- 57 Solve $x^3 - 6x^2 - 4x + 24 = 0$, given that $x = 6$ is a root. Find the other two roots as an ordered pair (smaller first).
- a. $(-2, 1)$
 - b. $(-2, 2)$
 - c. $(-1, 2)$
 - d. $(2, -2)$

- 58 The system $y = -x^2 - 7x - 3$ and $y = -2x + 3$ has two solutions. Find the sum of the x -coordinates of the two solutions.
- a. -5
 - b. -7
 - c. -3
 - d. 6
- 59 The system $y = -2x^2 - 13x - 32$ and $y = 5x + 8$ has two solutions. Find the sum of the x -coordinates of the two solutions.
- a. -11
 - b. 20
 - c. -9
 - d. -7
- 60 The system $y = -2x^2 - 7x - 8$ and $y = 5x - 8$ has two solutions. Find the sum of the x -coordinates of the two solutions.
- a. -8
 - b. 0
 - c. -4
 - d. -6
- 61 The system $y = -2x^2 + 14x - 22$ and $y = -2x - 8$ has two solutions. Find the sum of the x -coordinates of the two solutions.
- a. 7
 - b. 6
 - c. 10
 - d. 8

- 62 The system $y = -x^2 + 11x - 18$ and $y = 2x$ has two solutions. Find the sum of the x -coordinates of the two solutions.
- a. 11
 - b. 7
 - c. 9
 - d. 18